

The development of key science skills is a core component of the study of VCE Psychology and applies across Units 1 to 4 in all areas of study. These skills are used when planning and conducting your own research, as well as when analysing or evaluating the research of others (VCE Psychology Study Design June 2017 update, p. 11). All skills are evident in the different research methods commonly used by psychologists.

A **research method** is a particular way of conducting a research study ('investigation') to collect accurate and reliable data on a specific topic (question or problem) of interest. For example, experiments and self-reports are different types of research methods.

In an experiment, the researcher manipulates and controls a research participant's experiences in

some way to find out whether this causes a particular response. For example, a researcher may conduct an experiment to find out whether learning a list of previously unseen words by repeating the words and their definitions aloud three times improves performance on a test of those words. This could be compared with learning the words and their definitions by writing them down three times.

For some research topics, the most appropriate way of collecting data may be to ask participants about their thoughts, feelings or behaviour. This is when a self-report method such as a questionnaire or interview could be used. For example, participants may be asked questions about their attitudes towards school and the reasons for their attendance and absences. The questions may be presented in a questionnaire for which participants respond to a written set of questions, or by interviewing participants and recording their verbal answers in writing or electronically.

In some cases it is appropriate to use a combination of research methods to collect data. For example, a researcher conducting an experiment on different learning techniques used by students when studying for an exam may also interview the students to find out what motivates them to study for an exam, when they study and how much time they spend.

These are just some of the many research methods available to psychologists. Other methods include cross-sectional studies, case studies and observational studies.

In this chapter we examine the research methods and key science research skills prescribed in VCE Psychology. We start with an overview of the steps involved in conducting scientific research in psychology.

STEPS IN PSYCHOLOGICAL RESEARCH

Most of what psychologists know about behaviour and mental processes is based on empirical evidence. *Empirical evidence* is data collected through systematic observations and/or carefully controlled experiments. This type of evidence allows psychologists to draw accurate conclusions which are more likely to be valid and free from personal biases, as compared to our 'common sense' conclusions based on everyday observations of behaviour.

Because psychological research involves the collection of empirical evidence using scientific attitudes and practices, it is often called *empirical research*. Generally, a systematic step-by-step procedure is followed. There are variations of the steps and they do not guarantee that accurate and justifiable conclusions will be reached. However, the steps help ensure empirical evidence is collected and minimise the chance for bias, errors, faulty conclusions and results that cannot be tested through replication.

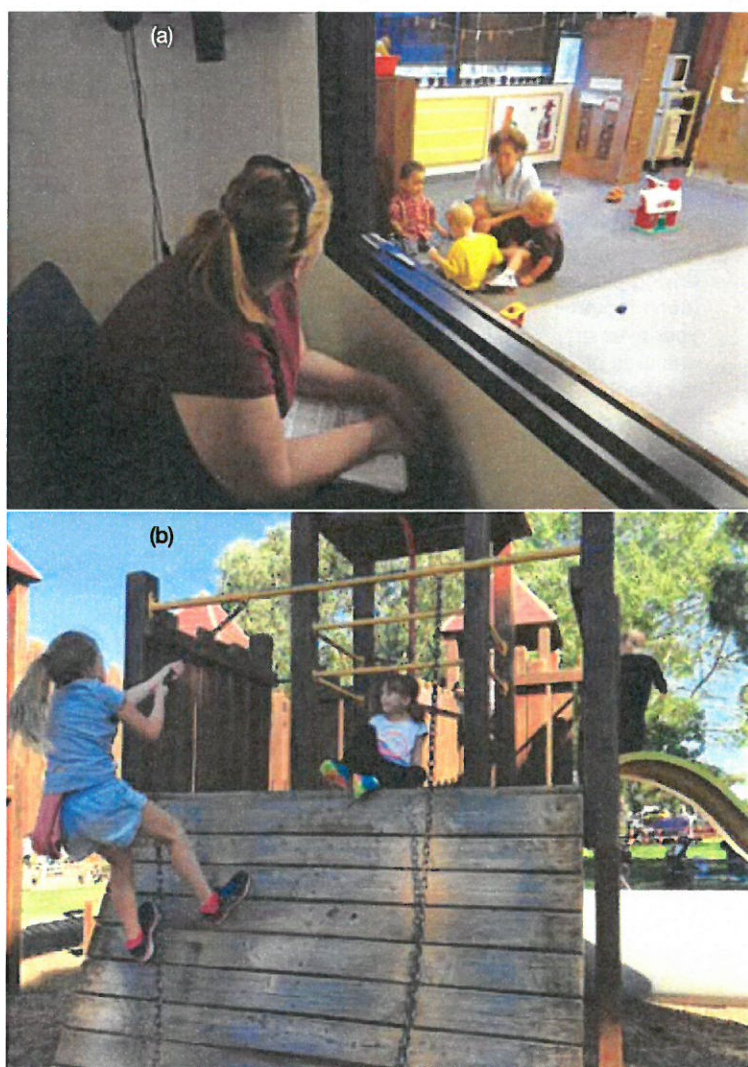


Figure 2.1 Experiments in psychology can be conducted in both laboratory and field settings. (a) In a laboratory setting, social behaviour may be observed in a controlled situation established by the researcher. (b) In a field setting, social behaviour may be observed in a real-world situation, but less control of conditions is possible.

Step 1: Identify the research topic

The first step in conducting psychological research is to identify the specific topic of research interest. This is often called the research 'question' or 'problem'. For example, a researcher might be interested in ways of reducing the number of accidents caused by red P-plate drivers.

To do this, they may conduct a *literature review*. This involves searching major psychological journals to find and review published reports of research studies that have already been undertaken on this topic. For example, they may consider research that has been conducted on defensive driving programs such as the Smith System.

The Smith System involves five rules to train the eyes to identify what is important when driving. As shown in Figure 2.3, the rules are:

1. Aim your vision high (to steer accurately and anticipate problems).
2. Keep your eyes moving (avoid staring and stay alert).
3. Look at the total driving picture (don't focus your eyes on one area of the road).
4. Look for and leave yourself a 'way out'.
5. Look for a position on the road that ensures other drivers can see you.

Conducting a literature review enables the researcher to become more familiar with their topic of research interest. It also enables them to refine their ideas and propose a relevant research question that will be the aim and therefore help focus their research activities; for example, 'Does training red P-plate drivers with the Smith System help to reduce the number of accidents they cause?'

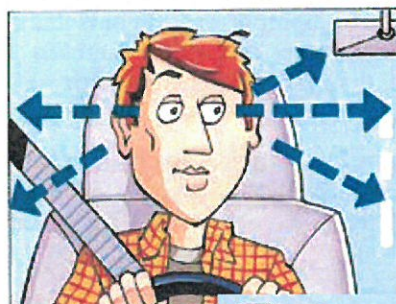


Figure 2.2 Psychological research may be conducted on ways of reducing the number of accidents caused by red P-plate drivers.

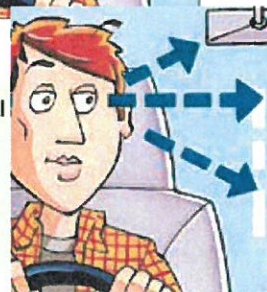
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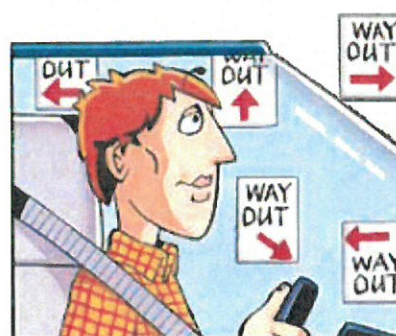
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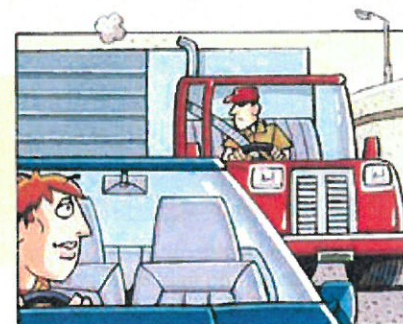


Figure 2.3 The Smith System involves five rules to identify what is important when driving.

Step 2: Formulate the research hypothesis

When the topic has been identified and refined, the next step is to formulate or construct a hypothesis for the research. This is essentially a thoughtful prediction about the results that will be obtained when the hypothesis is tested.

An example of a research hypothesis for the driver study is 'Red P-plate drivers who receive defensive driving training will make fewer driving errors than red P-plate drivers who have not received the training'.

The research hypothesis is formulated before the research study is conducted. Although there are various ways in which it can be expressed, it must be testable and written as a very specific statement.

The testability of the hypothesis for the Smith System study is reflected by the way in which it is stated. For example, 'number of accidents' is defined more specifically as 'driving errors' and driving errors will be measured through a practical driving test in a driving simulator. There are several other important characteristics of the research hypothesis. These are described on page 26.

Step 3: Design the research

The third step is to select the research method(s) and design the most appropriate procedures to collect the data required to test the hypothesis (but this is also considered when the hypothesis is being formulated). Generally this involves selecting a specific research method (such as a particular type of experiment) and modifying its various features to collect data that enables the hypothesis to be tested.

When designing the research, the researcher must consider the hypothesis, decide which participants will be studied, how many there will be, and how they will be selected and allocated to different groups that may be used in the study. The participants' responses provide the data that become the results for the research.

It is vital that relevant ethical standards and guidelines are followed to ensure the rights and wellbeing of all participants are protected. These apply to all stages of the research (including the report) and are described in the section starting on page 90.

There are also organisational matters to attend to; for example, ensuring there is access to a driving simulator, preparation of materials such as consent forms and instructions for participants and submitting the research for review and approval by an ethics committee.

Each type of research method that may be used to test a hypothesis has its advantages and limitations. Some are more suited to particular research hypotheses and data collection than others.

In the driver study, the researcher may decide to conduct an experiment using two separate groups (i.e. an independent groups design). Ten male and 10 female red P-plate drivers with a similar amount of driving

experience in city and regional areas as participants in the study could be used. Half the participants would be in one group and receive a defensive driving training session using the Smith System. The other half would be another group and not receive the training so that the potential benefits of the training could be compared. It would be important to randomly allocate all participants to the different groups to help ensure other participant characteristics that could have an effect on the results are evenly distributed across both groups. For example, some participants may have much more driving experience than others.

Alternatively, the researcher may decide to give questionnaires to a number of driving instructors who have taught young people, with and without the Smith System, to obtain information based on their experience as to which of the two methods resulted in better driving skills. Another option would be to conduct an experiment *and* interview the research participants.

Step 4: Collect the data

The fourth step involves actually collecting the required data in order to determine the results and conclusions that can be drawn.

Based on their plan and research design, the participants and materials are organised and the study is conducted.

Data may be *primary* or *secondary* and *quantitative* or *qualitative*. In the defensive driving experiment, the data will be primary as they are collected directly by the researcher through their experiment (rather than another source) and quantitative as they will be numerical (i.e. number of driving errors). These different types of data are more fully described in the section starting on page 70.

Step 5: Analyse the data

When the data have been collected they are in a 'raw' format because they have not been processed or analysed. The next step in psychological research is therefore to summarise, organise and represent the raw data in a logical and useful way to help determine whether the hypothesis is supported and to draw other conclusions.

This usually involves 'breaking down' a large set of numbers into smaller sets (e.g. raw data summarised as percentages in a table or graph) or even a single number or two (e.g. a mean score or standard deviation). *Descriptive statistics* are used to support this process. The researcher is then better able to consider the data when determining whether the hypothesis is supported based on the results obtained. Examples of descriptive statistics relevant to VCE Psychology are in the section starting on page 74.

Step 6: Interpret and evaluate the results

When the data have been analysed, they need to be interpreted and critically evaluated. This includes

drawing conclusions about what the results mean and identifying, describing and explaining limitations of the research and its results. The data may be considered as evidence and all conclusions must be considered as evidence-based. In much the same way as a crime investigator evaluates evidence, the researcher must consider the strengths and potential limitations of their evidence to help ensure their conclusions are consistent with the evidence and ultimately both valid ('accurate') and reliable ('stable').

One type of conclusion relates directly to the research hypothesis. The focus of this conclusion is

on whether the results will support the hypothesis. The researcher may also consider how widely the results of the research can be applied. *Inferential statistics* may be used by the researcher to help decide what the results mean and what conclusion(s) can legitimately be made. Examples of inferential statistics relevant to VCE Psychology are described in the section starting on page 84.

A researcher usually studies a relatively small number of participants who are selected from the bigger group of interest; for example, 10 male and 10 female red P-plate drivers aged between 18–21 years, rather than all red P-plate drivers. Of particular interest to the researcher



Figure 2.4 Steps commonly followed when planning and conducting psychological research. There are variations of these steps, but all are based on scientific attitudes and practices.

is whether the results obtained from a relatively small number of cases (the red P-plate drivers in the study) can be extended to apply to the bigger group of red P-plate drivers from which the smaller group was selected, to the entire group of red P-plate drivers, to green P-plate drivers or even all drivers.

With reference to the driving study, if the results for the experiment indicated that the red P-plate drivers who were trained with the Smith System made significantly fewer errors than the drivers who did not receive the training, then the researcher would interpret the results as providing support for the hypothesis. They would conclude that using the Smith System to teach defensive driving techniques to drivers reduces the likelihood of red P-plate drivers having an accident. On the basis of this conclusion, the researcher may also tentatively ('cautiously') conclude that the finding may apply to the bigger group of red P-plate drivers targeted for research or even all red P-plate drivers.

When drawing conclusions, the researcher bases their judgments strictly on what the results show. All conclusions must be consistent with the 'evidence' and relevant to the hypothesis that was tested.

They would seek to identify, comment on and take account of any limitations of their study, and its conclusions, particularly research procedures or unexpected events that may have influenced their results in an unwanted way.

They would also seek to identify further evidence that may be required, including how this could be obtained if the study were to be replicated.

Step 7: Report the research and findings

The final step of psychological research involves preparation of a report for others who may be interested in the research, its results and findings. Typically, psychological researchers prepare a detailed written report which they seek to get published in a professional journal. Each of these journals has reviewers who critically evaluate the research report and it is only on the basis of peer reviews that the research may be accepted for publication. In addition, a poster report may also be prepared for display and discussion at conferences or meetings with other researchers.

The report prepared for publication follows a strict format and describes in detail all aspects of the research, including relevant background information, how the research was conducted, the results and findings, implications of the findings, limitations which may have impacted on the results, recommendations, as appropriate, for modifying or extending the study, and a list of references used in preparing for the study

and writing the report. A poster report has a format more suited to display and is far less detailed.

Reporting the research and its findings is a very important part of the research process. It is the way other researchers find out about research which has been conducted and the way scientific progress is achieved. It also enables the general public to benefit from the findings of research.

Importantly, the reporting process places the specific study and its research procedures under the critical eye of other psychologists and researchers; for example, to check the accuracy of the findings and to consider limitations and alternative conclusions that may be valid. It also enables replication by other researchers – to repeat or reproduce the study in order to test the accuracy of the results or to test the relevance of the study or results to other groups or situations.

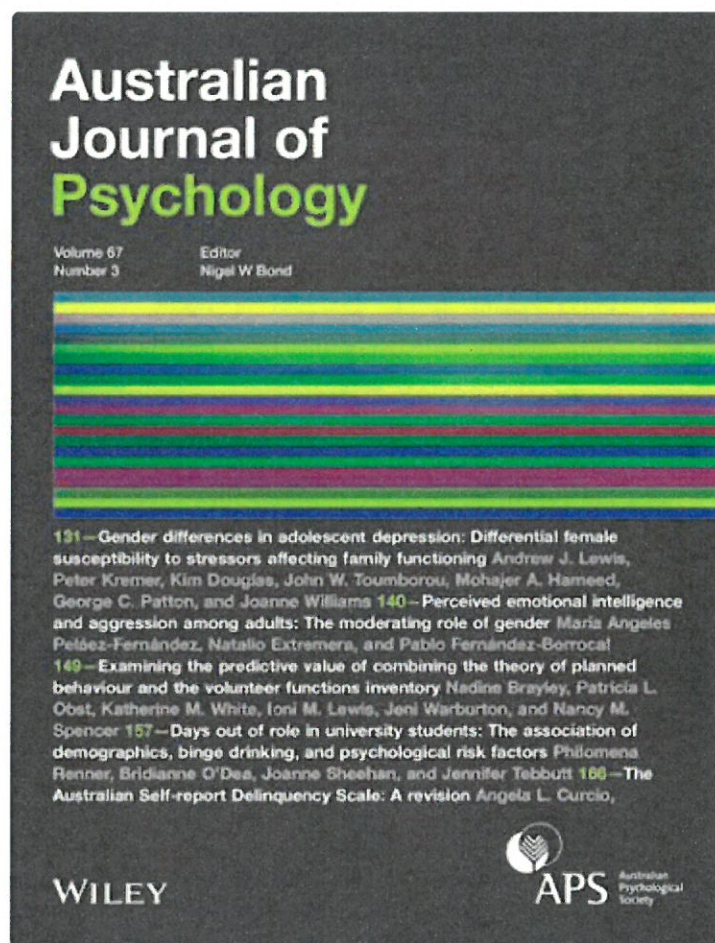


Figure 2.5 One of the Australian journals in which psychologists may publish their research reports

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View examples of journal articles

RESEARCH METHODS

Research methods are the means or 'tools' for observing, measuring, manipulating or controlling what takes place in psychological investigations. Each method has its specific purposes, procedures, advantages and limitations.

The choice of research method depends on which is most appropriate for the specific topic of research interest and hypothesis being tested. This is not unlike the choice made by a motor mechanic when selecting tools to repair a car engine. Their selection will depend on the specific engine problem in need of repair and the work that needs to be done to fix it. Each tool will have a specific use and way of being used. Similarly, each research method has a particular logic underlying its use and how it is used.

Sample selection and formulation of a research hypothesis are common to all psychological investigations requiring participants and are undertaken early in the research process. We consider each of these important procedures before examining the research methods prescribed in VCE Psychology and the specific features that distinguish them from each other.

Sample and population

Psychologists mostly conduct research studies with people and, in some cases, animals. The participants (or 'subjects') being studied are called the sample.

A **sample** is a subset or group that is part of a larger group chosen to be studied for research purposes. For example, suppose that a researcher is interested in conducting an experiment to find out whether children who attended a childcare centre during their preschool years have better language skills than children who did not attend a child care centre. It would be impractical to test every child who attended a child care centre and every preschool child who did not. Researchers would therefore select a sample with whom they conduct their investigation.

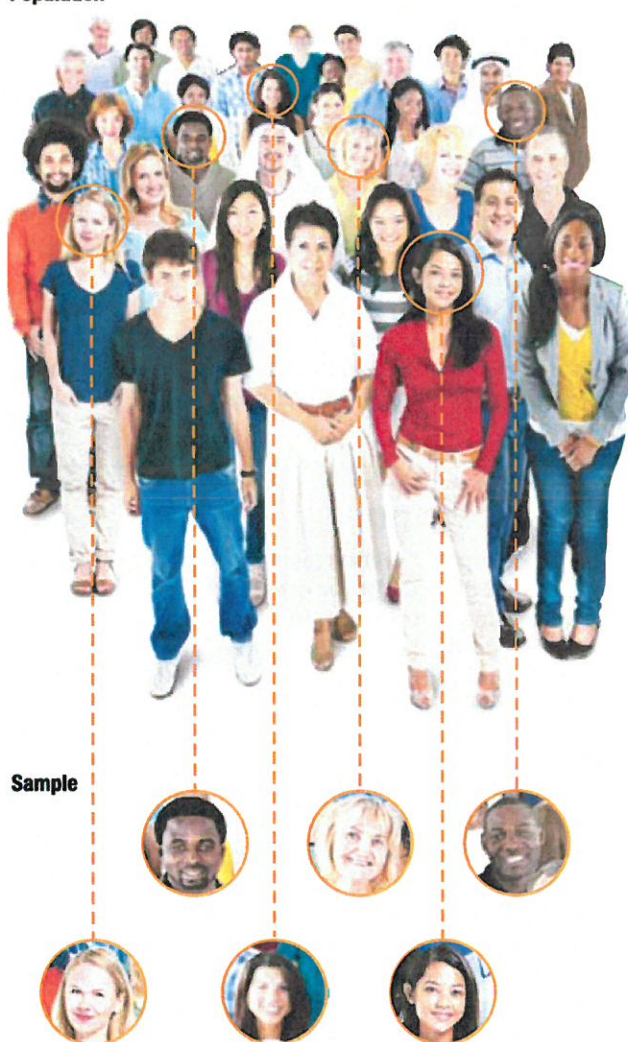
A sample is always smaller than the population. Therefore, an important feature of sample selection is that, whenever possible, the sample accurately reflect, or be representative of, the entire population of research interest. If this is achieved, the results obtained for the sample can be more confidently applied ('generalised') to the population from which it was drawn or even other groups or situations (assuming the results are valid and reliable).

In scientific research, the population does not necessarily refer to all people (or animals) in the world, in a country, or even in a particular city or area. The term **population** refers to the entire group of

research interest from which a sample is drawn and to which the researcher will seek to apply (generalise) the results of their investigation. A population of interest may be all preschool children, all blonde-haired females, all VCE Psychology students, all female VCE Psychology students, all Catholic school educated boys, or all male chimpanzees born in captivity.

Furthermore, a population used for research does not always refer to living things. A population could also be measurable things such as all community health centres in the Goulburn Valley region, all admissions at the Royal Melbourne Hospital, all VCE exam results in English in 2017, all days of school missed by year 9 students, all brands of sports shoes, all calls to the Kids Helpline telephone number, or any other specific source of data.

Population



Sample

Figure 2.6 This sample is a subset of the population of all staff who work for a multinational corporation.

LEARNING ACTIVITY 2.1

Review questions

1. Define the terms sample and population as they are used in research.
2. Draw a diagram to show the relationship between a sample and a population.
3. For the following research samples, identify two different populations from which each sample could be drawn.
 - (a) 20 year 10 girls and 20 year 10 boys
 - (b) 40 teachers who have been teaching for more than 10 years
 - (c) 100 employees on leave from work because of stress-related reasons
 - (d) 30 adults diagnosed as having an anxiety disorder
4. For the following research questions, identify a sample that might be used to conduct the research and a population from which the sample could be drawn.
 - (a) Are people who wear uniforms at school or work more likely to be obedient to an authority figure?
 - (b) How can people with a fear of flying be assisted to overcome their fear?
 - (c) Are children born to mothers aged over 40 years at greater risk of developing a mental disorder?
 - (d) Is it easier for men or women to give up smoking?

Research hypothesis

In psychology as in other sciences, different research methods are used to test one or more hypotheses relevant to the question or problem a researcher aims to study. A **research hypothesis** is a testable prediction of the relationship between two or more variables (events or characteristics). For example, it may be a prediction about the relationship between:

- attending a revision lecture (one event) and the score achieved for a psychology test (another event)
- biological sex (one characteristic) and finger dexterity (another characteristic)
- reading (one event) and brain wave activity (one characteristic).

The research hypothesis formulated for an investigation is essentially an educated or thoughtful guess about what the results will be. It is usually based on knowledge of other research findings or theories on the topic being studied. This is why it is often referred to as an 'educated' guess.

The research hypothesis is formulated before the study is conducted and guides the research. A useful research hypothesis typically has the following characteristics:

- refers to events or characteristics that can be observed and measured and is therefore testable (e.g. attendance at a revision lecture and what is learned are measurable and therefore data can be collected for hypothesis testing).
- states the existence of a relationship between two or more variables (e.g. a relationship between revision lecture attendance and scores on a psychology test)
- states the expected relationship between the variables, sometimes referred to as the 'direction of the relationship' (e.g. how revision lecture attendance will influence test scores)
- states a possible explanation of the results (e.g. higher psychology test scores will be attributable to revision lecture attendance)

- based on observations, a theory, model or research findings
- prepared as a carefully worded written statement (rather than a question)
- expressed clearly and precisely (rather than vaguely and generally)
- written as a single sentence.

In some cases, the research hypothesis may also refer to the population from which the sample was drawn and therefore the larger group about which the researcher intends to draw conclusions. The population, however, is most commonly described in the introduction to the report on the investigation.

Research hypotheses for the examples given earlier could be:

- Psychology students who attend a revision lecture before a test will achieve a better score on the test than students who do not.
- Females have better finger dexterity than males.
- Beta brain waves are predominant when reading.

There is no preferred writing style for a research hypothesis (nor is there a mandated style for VCE Psychology). Different writing styles can be equally valid. For example, some hypotheses use an 'if-then' style, such as 'if a certain event occurs, *then* it will cause a certain response'. In relation to the revision lecture attendance and test score 'events' (or variables), an if-then hypothesis may be stated as: 'If students attend a revision lecture, then they will achieve better test scores'. A more generally stated prediction such as 'Revision lecture attendance improves test results' would also be an appropriate alternative.

It is not always possible to be entirely certain about the accuracy of a prediction within a hypothesis. This is mainly because the researcher does not necessarily know or can control the influence of the many different variables that can affect the behaviour or mental process being studied. Nonetheless, many researchers would consider it pointless to conduct a study when the outcome is certain.

Research hypothesis versus theory and model

A research hypothesis is different from a theory and model. A research hypothesis is a specific prediction that guides the collection, analysis, interpretation and evaluation of data that has been collected to test it. In contrast, a **theory** or **model** is a general explanation of a set of observations about behaviour and/or mental processes which seem to be related.

The term model is used interchangeably with theory, however, many models in psychology tend to focus more on representing *how* some behaviour and/or mental process(es) could, should or does occur. For example, a model is often supported with or presented in the form of a diagram with boxes and arrows to organise and show relationships between different concepts. Figure 2.8 shows an example of a model used to represent and explain human memory in terms of inter-related sub-systems involving different types of memory called sensory memory, short-term memory and long-term memory. This model is also commonly referred to as a theory of memory.

Theories and models vary in scope, complexity and detail. All have one or more limitations. Some are essentially a hypothesis that has been restated. Others explain many interrelated research findings and ideas. Along with explaining existing results, a useful theory or model generates new hypotheses and guides further research.

Many theories or models of child development, personality, learning, remembering, forgetting and so on, are the products of psychological research and have generated valuable new research. In addition, some theories have generated new models and some models have generated new theories.

Whatever their scope — from tiny to vast — theories and models serve a gap-filling function. They explain how findings and ideas fit together in an organised way and what they mean, thereby making psychology a discipline that does more than report isolated facts.

Psychologists prefer testable theories and models because they can be confirmed, revised or rejected by further scientific research. Therefore, theories and models tend to not be judged in terms of their accuracy but rather in terms of their *usefulness*. This means that a theory or model tends not to be considered as right or wrong. Instead, it is simply regarded as more or less useful.

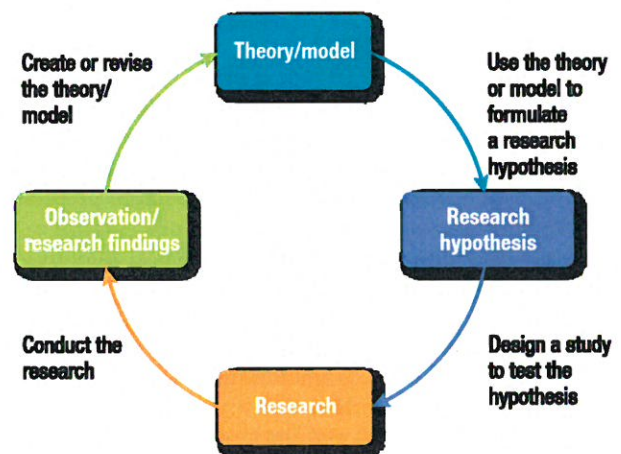


Figure 2.7 Theories and models are revised and expanded to reflect relevant research findings. New or revised theories and models lead to new observations or questions that stimulate new research.

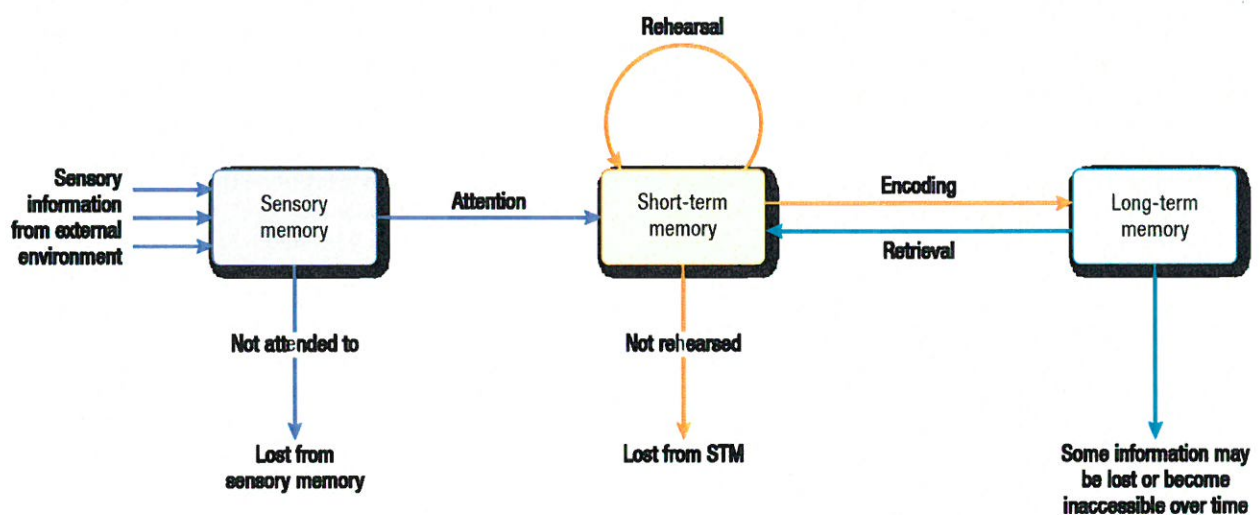


Figure 2.8 A contemporary model of human memory, representing it as a multi-storage system through which information flows